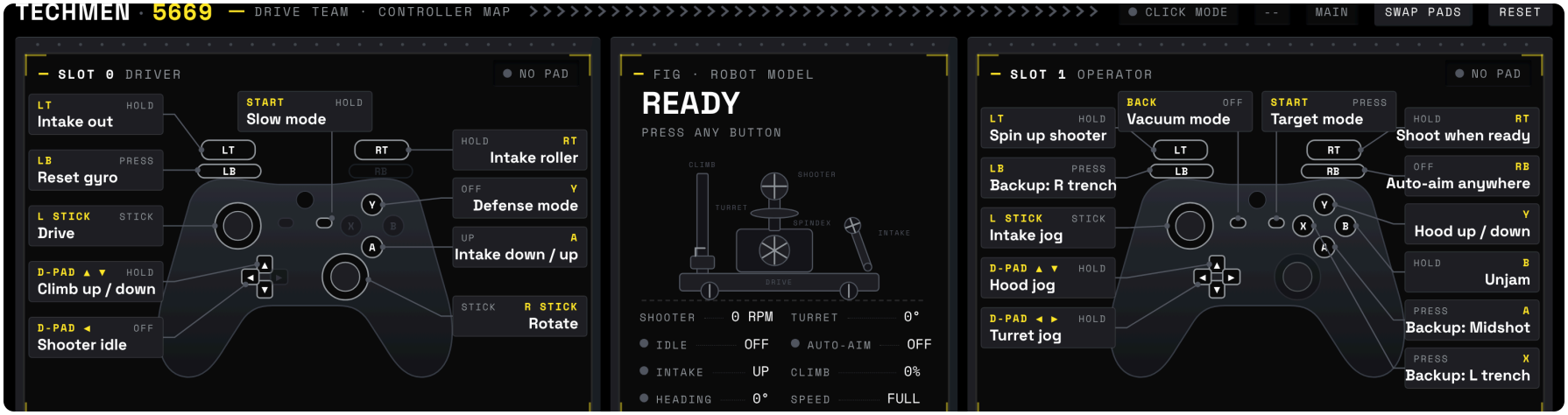
TECHMEN 5669 · Driver & Operator Controls

Robot code: main, Oct 26 2026 · Driver = Port 0 · Operator = Port 1



Driver · Port 0 · driving, intake, climb

Left Stick	STICK	Drive (forward, back, left, right)
Right Stick	STICK	Turn the robot
Start	HOLD	Slow mode
LB	PRESS	Reset gyro. **Point the robot straight away from your driver station first**
RT	HOLD	Intake roller: balls IN
LT	HOLD	Intake roller: balls OUT (spit them back)
A	PRESS	Intake down / up
D-pad Up / Down	HOLD	Climber up / down (stops by itself at the ends)
D-pad Left	PRESS	Shooter idle at 2000 RPM on / off
Y	PRESS	**Defense mode** on / off: intake up, hood down, quicker drive, wheels lock in an X when you let go
X	·	Nothing

Operator · Port 1 · aiming and shooting

RB	PRESS	**Auto-aim on / off** (turret, hood and shooter speed aim at our hub from anywhere)
RT	HOLD	Shoot. Auto-aim ON: fires only when ready (controller buzzes). OFF: feeds right away
LT	HOLD	Spin up the shooter (with auto-aim ON it spins up by itself while you hold RT)
Y	PRESS	Hood up / down
D-pad Up / Down	HOLD	Move the hood up / down slowly
D-pad Left / Right	HOLD	Turn the turret left / right
Left Stick up / down	HOLD	Move the intake up / down slowly (backup)
B	HOLD	Spindex backwards (unjam)
A / X / LB	PRESS	Backup shots: A = Midshot, X = Left trench, LB = Right trench
Back	PRESS	**Vacuum mode** on / off: intake + auto-aim, fires by itself (scores in our zone, passes from the neutral zone)
Start	PRESS	Target: AUTO → SCORE → PASS

Turret jog, hood jog or Y turns auto-aim OFF. Press RB to turn it back on.

TECHMEN 5669 · How To Use It

Keep this with the driver station

Before every match or practice

1. **Robot OFF.** Turret on the **HOME mark**. Hood **pushed all the way down** by hand. Intake **pushed all the way up**. Climber **all the way down**.
2. **Turn the robot on** while holding the hood down.
3. Driver Station **USB tab**: driver controller in **slot 0**, operator in **slot 1**. Press a button to see which row lights up; drag to fix. `F1` = rescan.
4. Driver Station **Team Station**: pick **our alliance color** (Red / Blue). At competition the field sets it.
5. Elastic: **"Pit Check"** should say **ALL GO**.
6. Enable, point the robot **straight away from your driver station**, press **Driver LB** once.
7. Shop field: **"Practice Field"** switch ON. **Competition: OFF.**

Emergency keys (Driver Station laptop)

- `Enter` = **Disable** (stops the robot, no restart needed). Use this one.
- `Space` = **Emergency stop**. The robot must be powered off and on again.

Shooting with auto-aim (normal way)

1. Operator presses **RB**: auto-aim ON. The turret and hood follow our hub by themselves.
2. Operator **holds RT**. The shooter spins up to the right speed for the distance and fires each ball the moment the shot is ready. The controller **buzzes** while it's ready.
3. The driver can keep driving. If the hub ends up behind the turret, the robot turns itself a little (the driver's turn stick always wins).

Shooting without auto-aim

1. Aim with **D-pad Left / Right** (turret) and **D-pad Up / Down** or **Y** (hood).
2. Hold **LT** to spin up, then **RT** to shoot. RT feeds right away, ready or not.

Backup shots (if the camera is having trouble)

1. Park on the spot: **Midshot** (between our hub and our climb), **Left trench** or **Right trench** (left / right as seen from our driver station).
2. Operator presses **A** (Midshot), **X** (Left trench) or **LB** (Right trench). This turns auto-aim on.
3. Hold **RT**.

Live version on any driver station laptop: **tools/controller-map/Start Controller Map.bat** (lights up each button as you press it). Full details: CLAUDE.md, "Controller Bindings".